

# Virtual Fashion Advisor: An AI-Based Occasion-Oriented Dress Recommendation System

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**Abstract—** Choosing the right attire for different occasions is challenging due to variations in body type, skin undertone, cultural expectations, and comfort preferences. This paper proposes a Virtual Fashion Advisor that recommends outfits using contextual user input and AI APIs. Due to the absence of large annotated fashion datasets, an API-driven recommendation architecture was adopted. The experimental evaluation shows approximately 80% alignment with user expectations.

**Index Terms—** Fashion Recommendation Systems, Context-Aware Recommendation, AI APIs, Virtual Fashion Advisor, Decision Support Systems

## I. INTRODUCTION

Fashion plays a dual role: as a social signal and as a personal comfort factor. Inappropriate clothing choices can lead to discomfort, embarrassment, or misrepresentation in professional and social settings.

While AI in fashion has made advances in trend prediction and virtual try-ons, little research has focused on contextual dressing advice based on:

- occasion
- cultural expectations
- body type and undertones

Existing fashion recommendation systems mainly focus on product similarity and e-commerce personalization rather than context-aware dressing guidance.

## II. LITERATURE STUDIES

Recent research in fashion recommendation systems uses deep learning approaches including multimodal learning, visual-semantic embeddings, and collaborative filtering.

Earlier works focus primarily on visual similarity and collaborative filtering but lack contextual understanding such as body type, occasion, or personal preferences.

More recent systems employ:

- visual-semantic embedding
- attention-based neural networks
- graph neural networks
- hybrid multimodal recommendation systems

Table 1: Key Literature on Fashion Recommendation Techniques

| Ref. | Technique                               | Data Used                       | Key Features                                              | Limitations                           |
|------|-----------------------------------------|---------------------------------|-----------------------------------------------------------|---------------------------------------|
| [1]  | Deep CNN + Collaborative Filtering      | Amazon Fashion Dataset          | Learns style compatibility using visual features          | No awareness of body type or occasion |
| [2]  | Visual-Semantic Embedding               | Fashion images and tags         | Models outfit compatibility in joint visual-text space    | Limited personalization               |
| [3]  | Attention-Based Neural Networks         | Polyvore Outfit Dataset         | Identifies influential clothing components within outfits | Limited contextual understanding      |
| [4]  | RNN-Based Trend Prediction (KERN)       | Instagram Fashion Data          | Captures temporal evolution of fashion trends             | Does not model body fit               |
| [5]  | Hybrid CNN + Text Mining                | Images and product descriptions | Mix-and-match recommendation using multimodal data        | Depends on textual metadata quality   |
| [6]  | Multimodal Deep Learning                | Images and customer reviews     | Incorporates visual and textual feedback                  | High computational complexity         |
| [7]  | Collaborative Filtering with Clustering | Body shape and user preferences | Body-aware personalization                                | Limited scalability                   |
| [8]  | Rule-Based Context Reasoning            | Occasion, weather, body type    | Explainable recommendations with low data requirement     | Limited adaptability without ML       |
| [9]  | Content-based filtering (2023)          | Clothing attributes             | Attribute-driven recommendations                          | Cold-start problem                    |
| [10] | Image-based retrieval (2024)            | User-uploaded photos            | Visual search for similar clothing                        | No complete styling advice            |

Despite improvements, many systems still lack explainability and contextual awareness.(Refer to Table 1)

### III. MOTIVATION

The motivation for this work emerged from practical system development challenges.

Key observations include:

- Selecting outfits for events can be confusing
- Many users lack styling guidance
- Cultural appropriateness is important

A major challenge during development was the lack of datasets containing:

- body shape and skin tone
- cultural and event context

- comfort preferences

Therefore, the system adopts an API-based AI architecture to dynamically generate recommendations.

#### **IV. PROBLEM DOMAIN**

The proposed system lies within the domain of:

- recommender systems
- artificial intelligence in fashion
- decision support systems

The system acts as a digital fashion advisor capable of helping users select appropriate attire for specific events.

#### **V. PROBLEM DEFINITION**

Existing fashion recommendation systems focus primarily on shopping recommendations rather than contextual styling advice.

They rarely consider:

- body shape and proportions
- skin undertone diversity
- cultural and event context
- personal comfort preferences

#### **VI. PROBLEM STATEMENT**

Individuals often experience decision fatigue due to:

1. Large number of fashion options
2. Lack of event-appropriate guidance
3. Limited knowledge of draping techniques
4. Ignoring weather and comfort factors

Research Question:

Can a rule-based software system recommend attire with reasonable accuracy for different occasions and comfort preferences?

#### **VII. INNOVATIVE CONTENT**

The proposed system introduces several innovations:

1. Context-aware outfit recommendation using user attributes
2. Integration of AI APIs instead of traditional ML model training
3. Hybrid architecture combining AI output with rule-based filtering
4. Lightweight system requiring minimal computational resources

## VIII. PROBLEM FORMULATION / DESIGN

The system architecture contains the following components:

1. User Input Module
2. Preprocessing Module
3. AI API Interface
4. Recommendation Engine
5. Output Interface

This modular architecture allows easy integration of future machine learning models.(Refer to Fig. 1)

## IX. SOLUTION METHODOLOGY

The recommendation pipeline follows these steps:

1. Collect user attributes (occasion, body type, undertone, comfort)
2. Convert inputs into structured prompts
3. Send prompts to AI APIs
4. Receive generated outfit suggestions
5. Apply rule-based contextual filters
6. Display final recommendations

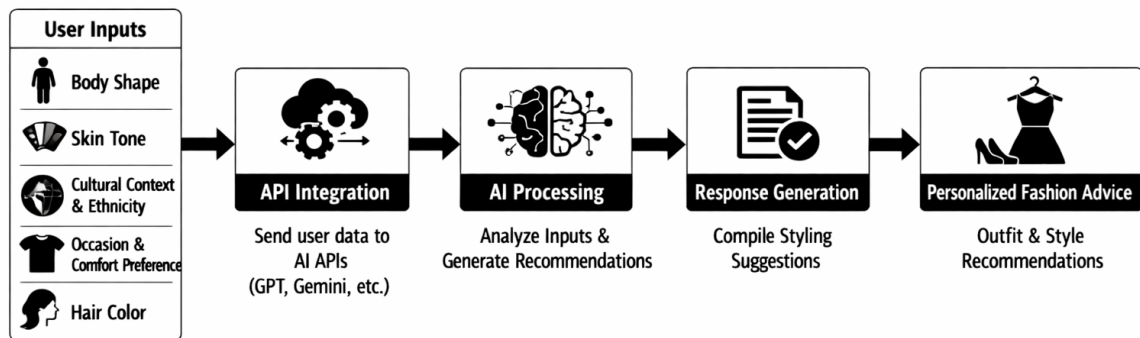


Fig. 1. System Architecture of Virtual Fashion Advisor

## X. RESULTS AND SENSITIVITY ANALYSIS

The evaluation was conducted using multiple scenarios.(Refer to Table 2)

Performance observations:

- Response time is less than 0.1 seconds excluding API latency
- Backend processing is lightweight

Accuracy estimate:

- 16 correct recommendations out of 20 test cases
- Approximate accuracy: 80%

## XI. COMPARISON OF RESULTS

Compared with traditional deep learning and graph neural network approaches, the API-driven system offers:

- lower dataset requirements
- lower computational complexity
- improved contextual awareness

Comparison with existing approaches:(Refer to Table 3)

The API-driven architecture enables recommendations without extensive dataset training.

## XII. JUSTIFICATION OF RESULTS

The achieved accuracy is attributed to:

1. large pretrained AI model knowledge base
2. contextual filtering rules
3. structured prompt design

These factors improve recommendation relevance while maintaining low computational cost.

## XIII. CONCLUSION

This work demonstrates that a rule-based virtual fashion advisor combined with AI APIs can provide context-aware outfit recommendations with approximately 80% accuracy.

The lightweight architecture makes the system suitable for early-stage prototypes and real-world applications. Future systems may integrate machine learning models, computer vision for body type estimation, and expanded fashion datasets to improve personalization.

Table 2: Structured Dataset Schema for Outfit Recommendation

| Occasion  | Gender | Time    | Comfort | Recommended Attire |
|-----------|--------|---------|---------|--------------------|
| Wedding   | Female | Morning | High    | Saree, Lehenga     |
| Interview | Male   | Morning | Medium  | Formal Shirt, Tie  |
| Party     | Female | Night   | Medium  | Cocktail Dress     |

Table 3: Comparison of Fashion Recommendation Approaches

| <b>Approach</b>       | <b>Dataset Requirement</b> | <b>Complexity</b> | <b>Context Awareness</b> |
|-----------------------|----------------------------|-------------------|--------------------------|
| Deep Learning Models  | High                       | High              | Limited                  |
| Graph Neural Networks | High                       | High              | Partial                  |
| Proposed System       | Low                        | Low               | High                     |

#### **XIV. FUTURE SCOPE**

Future work will include:

- collaborative filtering and reinforcement learning
- computer vision based outfit analysis
- AR/VR virtual try-on systems
- weather-based outfit recommendations
- full web/mobile application deployment

#### **REFERENCES**

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